

Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 5: Significance of Mitosis and Meiosis: Case Studies from Growth, Healing, Roots and Offspring

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
A	A	B	A	C	D	B

USE AFTER: OBSERVE PHASE — Jigsaw expert groups: Case B, a cut on the skin healing (boy cut by a panga while weeding shamba)

Q1. A boy is cut by a panga while weeding the shamba. After two weeks the skin has healed. Which statement best explains how the wound healed?

- A. The cells beside the cut divided by mitosis, making identical new skin cells to replace the damaged ones**
- B. Meiosis made new skin cells with half the chromosomes to fill the cut
- C. Meiosis made four different skin cells from each damaged cell
- D. No cell division happened; the old skin cells just stretched over the cut

Correct answer: A. Skin repair needs new body cells that are identical to the old ones with the full chromosome number, and this is what mitosis produces. Meiosis makes gametes, not skin cells.

USE AFTER: OBSERVE PHASE — Jigsaw expert groups: Case C, a plant producing new roots and shoots (sweet potato vine or sukuma wiki cutting)

Q2. A farmer cuts a piece of sweet potato vine and plants it. After some weeks it has grown new roots and shoots. Which type of cell division produced the new roots and shoots, and what is the evidence?

- A. Mitosis, because the new cells are identical to the parent cells and have the same chromosome number**
- B. Meiosis, because the new parts are made from gametes
- C. Meiosis, because plants need variation to grow roots
- D. Mitosis, because it halves the chromosome number in each new cell

Correct answer: A. Roots and shoots grow from new body cells that are identical copies with the full chromosome number. This is mitosis. No gametes or fertilisation are involved, and mitosis does not halve the chromosome number.

USE AFTER: OBSERVE PHASE — Jigsaw expert groups: Case E, parents producing children with different characteristics (siblings differing in height or skin tone)

Q3. Two brothers in the same family differ in height and skin tone, even though they have the same parents. Which explanation uses the correct cell division and evidence?

- A. Mitosis in the skin cells of each brother changes his height and skin tone
- B. Meiosis makes gametes that are genetically different from each other, so children differ from each other and from their parents**
- C. Mitosis makes gametes that are identical, so children copy one parent exactly
- D. Meiosis makes four identical gametes with a full set of chromosomes, so the children differ

Correct answer: B. Meiosis produces four genetically different haploid gametes, and this creates variation among offspring. Mitosis produces identical cells, so it cannot explain differences between siblings.

USE AFTER: OBSERVE PHASE — Case E teacher prompt: 'If both parents give a cell, how many chromosomes would a child have if each gave a full set? What must happen first?'

Q4. A human body cell has 46 chromosomes. Suppose gametes were made by mitosis instead of meiosis, so each sperm and each egg had 46 chromosomes. How many chromosomes would the fertilised egg have?

- A. 92
- B. 69
- C. 23
- D. 46

Correct answer: A. Mitosis keeps the chromosome number at 46, so the sperm (46) and egg (46) would give $46 + 46 = 92$ in the fertilised egg. The number would double each generation. This is why meiosis halves the number to 23 in each gamete.

USE AFTER: EXPLAIN PHASE — Think-pair-share: 'Using your table, explain how a 3 kg baby becomes a 50 kg teenager, and how each new cell gets the right instructions.'

Q5. A baby of 3 kg grows into a teenager of 50 kg. Which explanation is best supported by the evidence from the case studies?

- A. Food turns directly into new cells without any cell division
- B. The body cells only get bigger, and no new cells are made
- C. Mitosis makes many new identical cells using materials from food, and each new cell has the full set of chromosomes (instructions)**
- D. Meiosis in body cells makes the extra cells needed for growth

Correct answer: C. Growth needs a huge number of new body cells. Mitosis makes them from raw materials in food, and each daughter cell has the same full set of chromosomes, so it gets the right instructions. Cells getting bigger alone cannot explain it, and meiosis makes gametes.

USE AFTER: EXPLAIN PHASE — Discussion diamond: ranking the four claims about mitosis and meiosis

Q6. Which statement correctly compares the cells made by mitosis and by meiosis?

- A. Mitosis makes four different haploid cells; meiosis makes two identical diploid cells
- B. Mitosis makes gametes for reproduction; meiosis makes skin cells for healing
- C. Both make two identical cells, but meiosis is used only in plants
- D. Mitosis makes two identical diploid cells for growth and repair; meiosis makes four different haploid gametes**

Correct answer: D. Mitosis gives two genetically identical diploid daughter cells used for growth, repair and asexual reproduction. Meiosis gives four genetically different haploid gametes used in sexual reproduction.

USE AFTER: MODEL BUILDING PHASE — Revising the class model: a parent producing gametes by meiosis, with a fertilised egg restoring the full chromosome number ($23 + 23 = 46$)

Q7. On a class model, a parent cell with 46 chromosomes divides to make sperm cells with 23 chromosomes each. Which label and purpose should be written on the arrow?

- A. Mitosis: to make identical cells for growth
- B. Meiosis: to make gametes with half the chromosomes, so fertilisation restores 46 and the offspring show variation**
- C. Meiosis: to make body cells with 46 chromosomes for growth
- D. Mitosis: to repair damaged tissue

Correct answer: B. A change from 46 to 23 chromosomes in gametes is meiosis. Halving the number lets fertilisation restore 46 ($23 + 23$), and meiosis also creates variation. Mitosis keeps 46 in each new cell.